

EXEMPLAR-FREE ANALYTIC LEARNING FOR MULTI-LABEL AUDIO CLASS-INCREMENTAL LEARNING

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ABSTRACT

Audio classification is inherently a multi-label task, as real-world acoustic environments contain multiple simultaneous sound events. When new sound classes emerge, models must incorporate them without forgetting previously learned ones: a challenge known as class-incremental learning. Existing methods rely on storing past data and iterative gradient updates, which struggle under incomplete multi-label supervision because only the newly introduced classes are annotated at each phase, leaving old-class labels unavailable. We investigate exemplar-free analytic continual learning as a principled alternative, in which a linear classifier is updated in closed form without storing historical recordings or performing incremental back-propagation, and previously learned weights remain intact by construction. Building on analytic learning, we further propose ALMA, which addresses incomplete supervision and class imbalance through continuous old-class score estimates and frequency-based sample weighting. Experiments on a 50-class AudioSet-R benchmark across three incremental setups show that the analytic learner substantially outperforms gradient-based methods, and previously learned classes retain nearly unchanged detection performance as new classes are added. This study shows that ALMA is a simple yet effective solution to multi-label audio class-incremental learning.

Index Terms— Audio class-incremental learning, multi-label classification, analytic learning, catastrophic forgetting

1. INTRODUCTION

Sound recognition systems in smart homes, factories and ecological monitoring may face new sound classes after deployment [1–4], such as breaking glass, machine faults or newly observed wildlife calls. While single-label audio classification has advanced rapidly [5], it ignores the inherent nature of sound co-occurrence in real-world environments [6]. For instance, in a kitchen environment, sounds such as running water, sizzling, and speech naturally overlap simultaneously, a pattern that single-label models are inherently unable to capture. Learning sounds in isolation inevitably fails to represent these co-occurrence relationships. Multi-label audio classification emerges as a natural solution by predicting multiple sound events simultaneously [7]; however, the space of possible co-occurrence patterns is effectively unbounded in real world, necessitating that models continuously incorporate newly arriving sound classes without forgetting previously learned ones: a challenge referred to as multi-label audio class-incremental learning (CIL) [8].

Multi-label audio continual learning is especially challenging because each incremental phase provides labels only for its newly introduced class group [8, 9]. Suppose speech is learned at the first phase and vehicle sounds at the subsequent phase. A next-phase recording may contain both speech and a passing car, yet only the

vehicle classes are annotated. A missing speech label therefore does not indicate absence: the old-class annotation is simply unavailable. Training on only the annotated new classes thus pushes the model to actively suppress old-class predictions [10, 11], accelerating catastrophic forgetting [12, 13] rather than failing to reinforce it.

Continual learning methods mitigate catastrophic forgetting through replay [14], parameter regularization, or knowledge distillation [15–17], and recent work has extended these to multi-label CIL [10, 18, 19]. However, all rely on iterative gradient updates, which are particularly problematic in the multi-label ICL regime: missing annotations for old classes are interpreted as indicating absent classes, producing suppressive gradients that actively overwrite old knowledge. Knowledge distillation partially alleviates this, but the old model has never observed new-class sounds, making its targets unreliable precisely where the ambiguity is most severe.

Analytic continual learning instead derives the classifier in closed form via recursive least squares [20], and has been explored for single-label audio tasks including keyword spotting, sound-source localization, and audio deepfake detection [21–23]. Crucially, missing old-class labels contribute zero to the closed-form update, leaving old-class weights unchanged by construction: a property gradient-based methods cannot replicate. Whether this advantage extends to multi-label audio CIL, where labels cover only the current class group and leave old sound events unannotated, remains an open question.

To investigate this question, we propose an analytic framework for multi-label audio CIL that exploits the key property of the closed-form update leaving old-class weights intact by construction. Building on this, we further propose ALMA (Analytic Learning for Multi-label Audio), which extends the framework [11] with two mechanisms tailored to the multi-label regime. Rather than accepting zero-filled old-class targets, the previous-phase classifier generates soft predictions (i.e., pseudo-label) for old classes on current-phase recordings, providing a more faithful estimate of which old sounds are actually present. Frequency-based sample weighting is additionally incorporated to prevent dominant sound classes from skewing the classifier update [24]. To our knowledge, this is the first work to explore analytic continual learning for multi-label audio CIL. We also provide systematic evaluation to validate its effectiveness.

The findings show that analytic learning yields consistent improvements over gradient-based baselines, and ALMA with soft pseudo-labels and frequency weighting further demonstrates the benefit of explicitly addressing unannotated old classes. This suggests that the closed-form update is inherently better suited to the multi-label ICL regime than iterative gradient optimization, and that analytic continual learning is a promising direction for multi-label audio CIL in real-world deployments where exhaustive annotation across all sound classes is impractical¹.

¹The code will be released publicly.

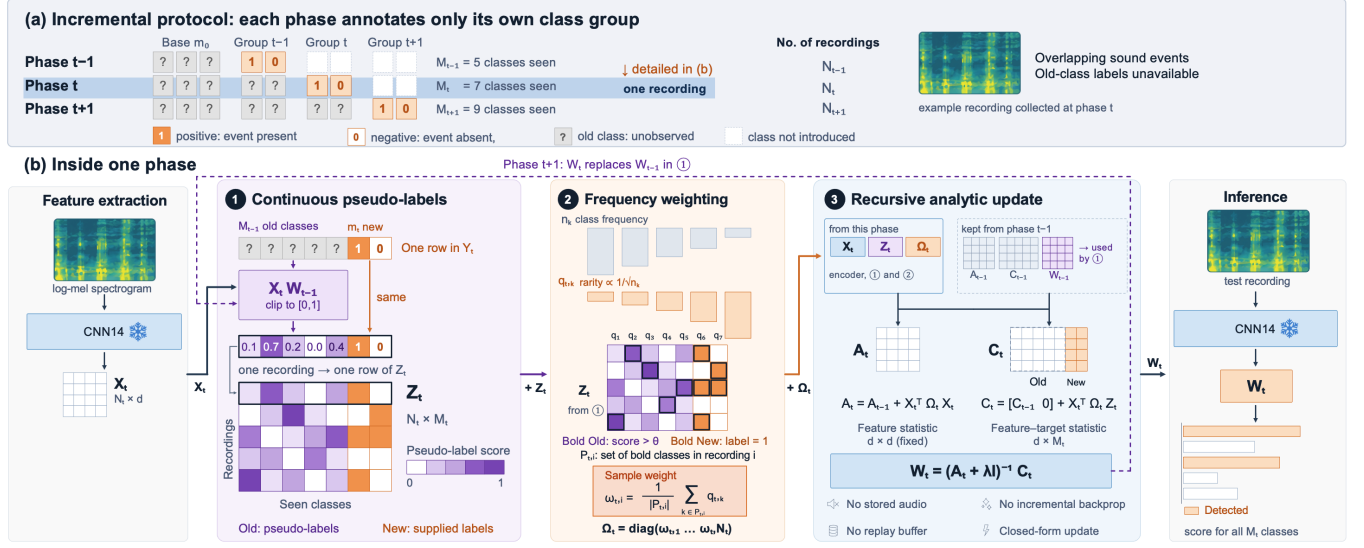


Fig. 1. Overview of ALMA. (a) At each phase only the new class group is annotated; old classes are unobserved, not negative. (b) Within phase t , a frozen CNN14 gives X_t ; (1) clipped scores $X_t W_{t-1}$ and Y_t complete the target matrix Z_t ; (2) rarity-based weights form Ω_t ; (3) X_t, Z_t, Ω_t update A_t, C_t recursively, giving W_t in closed form and returning it to step (1) at the next phase. Inference scores all M_t classes with the frozen encoder and W_t .

2. ALMA FOR AUDIO MULTI-LABEL CIL

2.1. Problem Setting and Notation

We follow the established multi-label class-incremental learning protocol [8, 9, 11], in which classes are introduced over a sequence of phases. At phase t , training annotations are provided only for the class group introduced in that phase. However, because an audio recording can contain several overlapping sound events, a current recording may also contain events from previously learned classes. Their annotations are unavailable during the current phase.

Formally, as shown in Figure 1(a), the learning process unfolds over phases $t = 0, 1, \dots, T$. At each phase t , a new group of m_t sound classes is introduced, and the learner has seen $M_t = \sum_{i=0}^t m_i$ classes in total by the end of that phase. At phase t , the learner receives N_t audio recordings. Each recording is passed through a frozen audio encoder to produce a d -dimensional feature vector, and the full set of recordings is represented as a feature matrix $X_t \in \mathbb{R}^{N_t \times d}$ where each row corresponds to one recording.

The annotations at phase t cover only the m_t newly introduced classes, forming a binary label matrix $Y_t \in \{0, 1\}^{N_t \times m_t}$ where 1 denotes a positive label indicating the presence of a newly introduced sound class, and 0 denotes a negative label indicating its absence. Y_t has no columns for the M_{t-1} classes learned in previous phases.

To train a classifier over all M_t classes seen so far, we construct a completed target matrix $Z_t \in [0, 1]^{N_t \times M_t}$ whose columns are assembled from two sources: the current annotations Y_t fill the columns for new classes, while the classifier from phase $t-1$ supplies soft score estimates for the old-class columns. The analytic classifier $W_t \in \mathbb{R}^{d \times M_t}$ is then fitted to map X_t to Z_t in closed form, and produces a score for each of the M_t classes for any input.

2.2. Analytic Learning

As shown in Figure 1(b) (left), given N_t audio recordings at phase t , the encoder f_θ extracts a d -dimensional feature vector $x_i = f_\theta(a_i)$ for each recording a_i , stacked into a feature matrix $X_t \in \mathbb{R}^{N_t \times d}$. A

linear classifier $W_t \in \mathbb{R}^{d \times M_t}$ maps these features to scores over all M_t sound classes seen so far. After the base phase, the encoder is frozen to preserve the learned acoustic representations across phases, while the linear classifier is updated analytically using X_t and the completed target matrix Z_t to incorporate newly introduced sound classes without gradient updates or access to historical recordings. Here, the old classes in Z_t not annotated are represented as 0.

The classifier W_t is estimated by minimizing the ridge regression objective

$$\min_{W_t} \|X_t W_t - Z_t\|_F^2 + \lambda \|W_t\|_F^2, \quad (1)$$

where the first term minimizes prediction error over all M_t classes and $\lambda > 0$ controls the regularization strength. This objective admits the closed-form solution

$$W_t = (X_t^T X_t + \lambda I_d)^{-1} X_t^T Z_t. \quad (2)$$

The solution depends on the training data only through two sufficient statistics

$$A_t = X_t^T X_t \in \mathbb{R}^{d \times d}, \quad C_t = X_t^T Z_t \in \mathbb{R}^{d \times M_t}, \quad (3)$$

so that $W_t = (A_t + \lambda I_d)^{-1} C_t$. When phase t arrives, these statistics are accumulated as

$$A_t \leftarrow A_{t-1} + X_t^T X_t, \quad C_t \leftarrow C_{t-1} + X_t^T Z_t, \quad (4)$$

and W_t is recomputed from the updated statistics via Eq. (2). This recursive update is mathematically equivalent to retraining jointly on all phases seen so far, yet requires neither historical recordings nor back-propagation [20].

2.3. Proposed ALMA

Figure 1(b) presents ALMA. Building on the analytic learning framework described above, ALMA incorporates two mechanisms to address the missing-label regime of multi-label audio CIL: continuous pseudo-labeling and frequency-based sample weighting.

2.3.1. Continuous Pseudo-Labels

At phase $t \geq 1$, the supplied label matrix $\mathbf{Y}_t$ covers only the m_t newly introduced classes, leaving old-class labels unobserved. Rather than filling these with zeros, we use the preceding classifier $\mathbf{W}_{t-1}$ to estimate them. Specifically, we compute old-class regression scores

$$\mathbf{S}_t^{\text{old}} = \mathbf{X}_t \mathbf{W}_{t-1} \in \mathbb{R}^{N_t \times M_{t-1}}, \quad (5)$$

and clip each entry to $[0, 1]$ to obtain soft pseudo-labels $\tilde{\mathbf{Y}}_t^{\text{old}}$. Retaining continuous scores rather than thresholding them preserves uncertainty about which old sound classes are present. The completed target matrix is then

$$\mathbf{Z}_t = \left[\tilde{\mathbf{Y}}_t^{\text{old}} \mid \mathbf{Y}_t \right] \in [0, 1]^{N_t \times M_t}, \quad (6)$$

where each row contains soft targets for old classes followed by binary targets for the current class group. At the base phase, $\mathbf{Z}_0 = \mathbf{Y}_0$.

2.3.2. Frequency-Based Sample Weighting

Audio events are generally imbalanced, with some events occurring more often and others occurring less frequently. To prevent dominant classes from skewing the classifier update, each recording is assigned a weight inversely proportional to the frequency of its positive classes.

Concretely, each recording i is assigned a scalar weight $\omega_{t,i}$ that reflects the rarity of its positive sound classes. Let n_k denote the number of positive recordings for class k at its introduction phase. The rarity factor of class k is defined as

$$q_{t,k} = M_t n_k^{-1/2} / \sum_{\ell=1}^{M_t} n_\ell^{-1/2}, \quad (7)$$

which assigns larger values to less frequent classes, and then normalizes the mean rarity factor across all M_t classes to one [11, 24]. The weight of recording i is the average rarity factor over its positive classes $P_{t,i}$, which includes both supplied new-class labels and pseudo-labeled old classes whose scores exceed a threshold θ :

$$\omega_{t,i} = \frac{1}{|P_{t,i}|} \sum_{k \in P_{t,i}} q_{t,k}. \quad (8)$$

These per-recording weights are collected into a diagonal matrix, $\mathbf{\Omega}_t = \text{diag}(\omega_{t,1}, \dots, \omega_{t,N_t})$, and used in the closed-form solution.

2.3.3. Recursive Analytic Update

The completed targets $\mathbf{Z}_t$ and sample weights $\mathbf{\Omega}_t$ together define the weighted ridge regression objective accumulated over all phases:

$$\mathcal{J}_t(\mathbf{W}) = \sum_{\tau=0}^t \left\| \mathbf{\Omega}_\tau^{1/2} (\mathbf{X}_\tau \mathbf{W} - \mathbf{Z}_{\tau \rightarrow t}) \right\|_F^2 + \lambda \|\mathbf{W}\|_F^2, \quad (9)$$

where $\mathbf{Z}_{\tau \rightarrow t}$ pads the targets from phase τ with zero columns for classes introduced in later phases. Rather than storing the feature and target matrices from every phase, the solution to Eq. (9) can be recovered from two sufficient statistics. At the base phase, these are now initialized as

$$\mathbf{A}_0 = \mathbf{X}_0^\top \mathbf{\Omega}_0 \mathbf{X}_0, \quad \mathbf{C}_0 = \mathbf{X}_0^\top \mathbf{\Omega}_0 \mathbf{Z}_0. \quad (10)$$

At each subsequent phase, they are updated as

$$\mathbf{A}_t = \mathbf{A}_{t-1} + \mathbf{X}_t^\top \mathbf{\Omega}_t \mathbf{X}_t, \quad (11)$$

$$\mathbf{C}_t = [\mathbf{C}_{t-1} \quad \mathbf{0}_{d \times m_t}] + \mathbf{X}_t^\top \mathbf{\Omega}_t \mathbf{Z}_t, \quad (12)$$

where $[\mathbf{C}_{t-1} \quad \mathbf{0}_{d \times m_t}]$ denotes the horizontal concatenation of $\mathbf{C}_{t-1}$ and a zero matrix containing one column for each newly introduced class. The retained $\mathbf{C}_{t-1}$ preserves all prior supervision for old classes, while the current term adds contributions from phase- t recordings. The classifier is then recovered in closed form as

$$\mathbf{W}_t = (\mathbf{A}_t + \lambda \mathbf{I}_d)^{-1} \mathbf{C}_t. \quad (13)$$

Here, the weight is introduced and used to update the solution.

3. EVALUATION PROTOCOL

3.1. Data and baselines

We use a 50-class subset of AudioSet-R [25], selecting the 50 most frequent classes ordered from most to least frequent. Each ten-second clip may carry multiple positive labels and can appear in more than one phase set. Training at each phase exposes only the current class group's labels, while evaluation uses all labels for classes seen so far.

Audio is represented by 64-bin log-mel spectrograms. We train CNN14 [26] during the base phase and freeze the encoder thereafter. Its 4096-dimensional embeddings provide the fixed features.

Baselines include fine-tuning (FT), elastic weight consolidation (EWC), synaptic intelligence (SI), learning without forgetting (LwF), Joint training, and per-phase retraining (PPR) [15–17]. We adapt these gradient-based methods to multi-label audio using independent per-class sigmoid outputs and binary cross-entropy. FT uses only current-phase clips and serves as a lower-bound reference. Joint and PPR access historical recordings, while EWC, SI, and LwF are regularization and distillation baselines. Hyperparameters are selected on a separate validation set. Average precision (AP) measures the ranking of positive clips for one class, and mean average precision (mAP) averages AP across classes. Cumulative mAP covers all classes observed at each phase. Mean cum. averages these phase scores. Final mAP is the last-phase score.

4. RESULTS

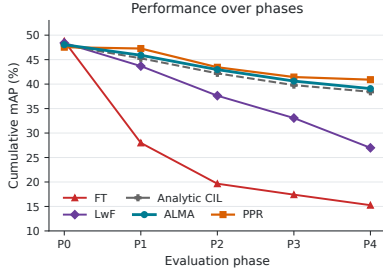
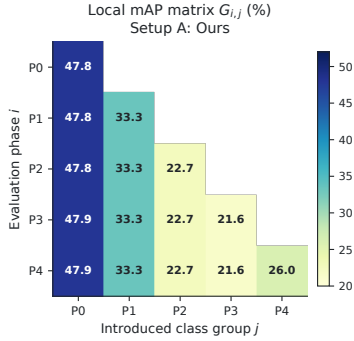
4.1. Overall Incremental Performance

Table 1 summarizes the results. Analytic CIL, which uses zero-filled old-class targets without $\mathbf{\Omega}$, already achieves 42.73% mean cumulative mAP and 38.43% final mAP, outperforming all gradient-based methods. The strongest of these, LwF, obtains 37.95% and 26.98%, respectively. ALMA further raises these by 0.59 and 0.65 percentage points to 43.32% and 39.08%. The much larger gap between gradient-based methods and Analytic CIL shows that the analytic framework shows benefits in this task, and the pseudo-labeling and frequency weighting components provide additional gain.

The strong performance of the analytic methods can be attributed to how they incorporate knowledge from previous phases. Gradient-based methods optimize only on current-phase data at each update, which risks overwriting what was learned before. The analytic framework instead accumulates all previous phases exactly into a single closed-form solution, so no previously learned information is discarded. This is further supported by the comparison with PPR, which retraining on all historical data at every phase and serves as an upper bound. ALMA closes most of this gap, scoring only 0.80 and 1.82 points below PPR despite storing no historical recordings. As shown in Figure 2, ALMA starts below FT and LwF at the base phase but surpasses both after the first incremental update, confirming that its advantage grows as more phases are observed.

Table 1. Performance comparison with baselines.

Method	Past data	Inc. BP	Mean mAP	Final mAP
FT	✗	✓	25.82	15.26
EWC [15]	✗	✓	25.91	14.96
SI [16]	✗	✓	33.22	25.25
LwF [17]	✗	✓	37.95	26.98
Analytic CIL (Ours)	✗	✗	42.73	38.43
ALMA (Ours)	✗	✗	43.32	39.08
Joint	✓	—	43.46	40.45
PPR	✓	✓	44.12	40.90

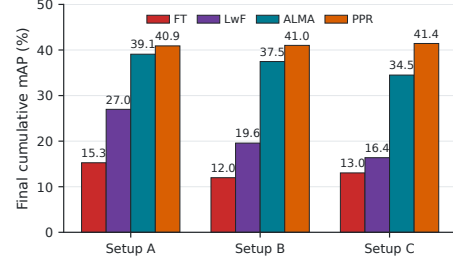
**Fig. 2.** Cumulative mAP across evaluation phases. The comparison shows how the advantage of the analytic methods emerges as new class groups are introduced.**Fig. 3.** Local-mAP matrix $G_{i,j}$ on a fixed test pool. Row i is the evaluation phase and column j the class group; blanks are groups not yet introduced.

4.2. New-Class Learning and Old-Class Retention

Figure 3 presents the local mAP of each class group across phases, where each column tracks one group from its introduction onward. If the model forgets previously learned classes as new ones are added, local mAP would decline across a column. Instead, the base group (P0) maintains approximately 47.8% local mAP through all subsequent phases, and the group introduced at P1 similarly holds at approximately 33.3%. This stability across all class groups demonstrates that ALMA retains previously learned classes as new ones added, confirming strong resistance to catastrophic forgetting.

4.3. Performance Across Incremental Protocols

Figure 4 evaluates whether the advantage of ALMA persists as the base phase shrinks and the number of incremental phases grows. Setup A begins with 30 base classes followed by four five-class phases; Setup B with 20 followed by six; and Setup C with 10 followed by eight: providing a progressively more challenging evaluation. ALMA achieves final mAP of 39.08%, 37.47%, and 34.50% across Setups A–C, consistently outperforming LwF by 12.10,

**Fig. 4.** Final cumulative mAP (%) across Setups A–C with increasingly long incremental sequences.**Table 2.** Setup A ablation. PL denotes pseudo-labels; Mean averages P0–P4.

Variant	Old-class target	Ω	P0	P1	P2	P3	P4	Mean
Analytic CIL	Zero-filled	✗	47.968	45.270	42.203	39.791	38.429	42.732
	Zero-filled	✓	48.045	44.702	41.528	39.086	37.862	42.245
	Hard PL	✗	47.968	45.639	42.638	40.178	38.706	43.026
	Hard PL	✓	48.045	45.705	42.659	40.177	38.700	43.057
	Continuous PL	✗	47.968	45.829	42.943	40.622	39.111	43.295
ALMA	Continuous PL	✓	48.045	45.905	42.962	40.619	39.084	43.323

17.89, and 18.13 points and FT by 23.82, 25.49, and 21.47 points. It also remains close to PPR, the historical-data upper bound, with gaps of only 1.82, 3.54, and 6.93 points. The widening gap to PPR reflects the increasing difficulty of later setups, but the advantage over gradient-based methods remains consistent throughout.

4.4. Ablation Study

Table 2 presents a matched ablation isolating the contributions of old-class target construction and sample weighting beyond the basic analytic learner. Hard PL denotes binarizing the preceding classifier’s old-class scores at a threshold of 0.5, whereas Continuous PL retains the clipped scores in $[0, 1]$. Replacing zero-filled old-class targets with hard pseudo-labels improves mean cumulative and final mAP by 0.29 and 0.28 points, and retaining continuous scores instead of hard labels adds a further 0.27 and 0.40 points. Adding sample weighting Ω to continuous targets changes mean and final mAP by +0.03 and −0.03 points respectively. These results show that old-class target construction provides the main additional gain beyond the basic analytic learner, with a modest advantage from continuous over hard pseudo-labels, while sample weighting provides no consistent additional benefit.

5. CONCLUSION

We investigated analytic continual learning for multi-label audio CIL without historical recordings or incremental back-propagation. The analytic learner substantially outperforms gradient-based methods across all evaluated setups, identifying the closed-form recursive update as the main source of improvement. ALMA further improves performance by replacing zero-filled targets with continuous old-class score estimates, demonstrating that better target construction consistently benefits the analytic framework. Local mAP of previously learned class groups remains nearly unchanged as new classes are added, confirming strong resistance to catastrophic forgetting. These results establish analytic learning as an effective and computationally simple direction for multi-label audio CIL.

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